ព្រះរាជាណាចក្រកម្ពុជា

ជាតិ សាសនា ព្រះមហាក្សត្រ



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Project: Web Scaping & Data Analysis (Reddit Data Science)

Subject: Programming for Data Science

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1.Abstract

Social media platforms generate large volumes of user-created content that can be analyzed to understand online behavior and community interests. Reddit is a popular discussion platform organized into topic-based communities known as subreddits. One such community, *r/datascience*, serves as a space for discussions related to data science concepts, tools, and career development.

This project focuses on collecting and analyzing data from the *r/datascience* subreddit using web scraping and exploratory data analysis (EDA) techniques. A dataset consisting of 1,100 Reddit posts was collected using Python libraries including Requests, BeautifulSoup, and Pandas. The dataset contains information such as post titles, scores, number of comments, authors, timestamps, and flairs. Several data preprocessing steps were applied, including handling missing values, converting timestamps, and creating an engagement metric based on post scores and comments.

Exploratory data analysis was conducted to examine posting trends over time, engagement distribution, and topic popularity. The results show that user engagement is highly skewed, with a small number of posts receiving a large portion of interactions. Discussion-based and career-related posts were found to generate higher engagement compared to other topics. This study demonstrates how web scraping combined with EDA can be used to extract meaningful insights from online communities and highlights the potential of social media data for behavioral and trend analysis.

2. Introduction

With the increasing availability of online data, social media platforms have become an important source for data analysis and research. Reddit is one of the largest discussion-based platforms, where users create and participate in topic-focused communities called subreddits. These communities generate large amounts of unstructured data in the form of posts, comments, and user interactions, making Reddit a valuable platform for applying data science techniques.

The *r/datascience* subreddit is a highly active community where users discuss topics such as data analysis methods, machine learning, programming tools, academic learning, and career guidance. Due to the volume and frequency of posts, manually analyzing discussions to understand trends and engagement patterns is inefficient. Therefore, automated data collection and analysis techniques are required to extract meaningful insights.

In this project, Python is used to perform web scraping and exploratory data analysis (EDA) on posts collected from the *r/datascience* subreddit. The implementation is carried out in a Jupyter Notebook environment using libraries such as **Requests** and **BeautifulSoup** for scraping, and **Pandas**, **NumPy**, and **Matplotlib/Seaborn** for data cleaning, analysis, and visualization. The

scraping process extracts key attributes including post titles, scores, number of comments, authors, timestamps, and flairs, which are then stored in a CSV file for further analysis.

Problem Statement

1.How does the number of posts in the *r/datascience* subreddit change over time?

2.How do users engage with posts in terms of scores and comment counts?

3.How are comment counts distributed, and do they show skewness or outliers?

4.Is there a relationship between the number of upvotes (score) and the number of comments on a post?

5.How does user engagement vary across different post flairs?

Project Objectives

The objective of this project are to:

- To scrape Reddit posts from the *r/datascience* subreddit
- To clean and preprocess the scraped data
- To analyze posting trends over time
- To study engagement patterns using scores and comments
- To identify popular topics and flairs
- To visualize insights using appropriate charts

3. Data Collection (Web Scraping)

This section describes the data source, tools, scraping methodology, and ethical considerations used to collect Reddit data for this project.

3.1 Data Source

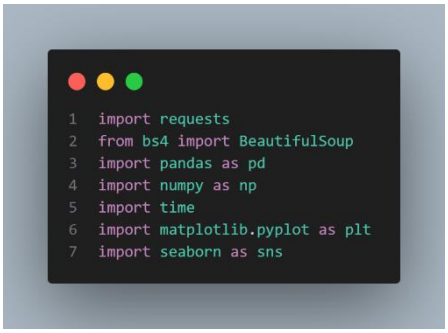
The data for this study was collected from the public subreddit **r/datascience** on the Reddit platform. This subreddit is dedicated to discussions related to data science concepts, tools, methodologies, and career development. The subreddit was selected due to its high activity level and relevance to the objectives of this project. Data was accessed through Reddit's publicly available web pages using the old.reddit.com interface, which provides a consistent HTML structure suitable for web scraping.

Source from Website : https://old.reddit.com/r/datascience

3.2 Tools and Technologies

The web scraping and data collection process was implemented using the Python programming language. The following libraries were used:

- **Requests** for sending HTTP requests and retrieving webpage content
- **BeautifulSoup** for parsing HTML documents and extracting relevant information
- **Pandas** for storing, cleaning, and managing the collected data
- **NumPy** for basic numerical operations



```python
import requests
from bs4 import BeautifulSoup
import pandas as pd
import numpy as np
import time
import matplotlib.pyplot as plt
import seaborn as sns
```

These tools enabled efficient data extraction, processing, and storage in CSV format for further analysis.

3.3 Scraping Process

The scraping process involved sending HTTP requests to multiple pages of the *r/datascience* subreddit and parsing the HTML content using BeautifulSoup. For each Reddit post, key attributes such as the post title, score, number of comments, author username, timestamp, and flair were extracted. Pagination was handled to collect a sufficient number of posts, resulting in a dataset of 1,083 records. The extracted data was structured into a Pandas DataFrame and saved as a CSV file for subsequent preprocessing and analysis.



```python
for post in posts:
        try:
            # Extract data using the selectors from your project
            title = post.find("a", class_="title").text if post.find("a", class_="title") else "N/A"
            score = post.find("div", class_="score unvoted").get("title") if post.find("div", class_="score unvoted") else 0
            comments = post.find("a", class_="comments").text.split()[0] if post.find("a", class_="comments") else 0
            author = post.find("a", class_="author").text if post.find("a", class_="author") else "N/A"
            time_posted = post.find("time")["datetime"] if post.find("time") else None
            flair = post.find("span", class_="linkflairlabel").text if post.find("span", class_="linkflairlabel") else "No Flair"

            posts_data.append({
                "Title": title,
                "Score": int(score) if str(score).isdigit() else 0,
                "Comments": int(comments) if str(comments).isdigit() else 0,
                "Author": author,
                "Time": time_posted,
                "Flair": flair
            })
```

3.4 Ethical Considerations

Ethical guidelines were followed throughout the data collection process. Only publicly available data was collected, and no private or sensitive user information was accessed. The scraping frequency was controlled to avoid excessive requests to Reddit servers. This study adheres to responsible data collection practices and respects the platform's terms of use.

4.Dataset Descriptions

4.1 Overview of the Dataset

	Title	Score	Comments	Author	Time	Flair
0	Weekly Entering & Transitioning - Thread 26 Ja...	16	10	AutoModerator	2026-01-26T05:01:28+00:00	No Flair
1	[Official] 2025 End of Year Salary Sharing thread	115	141	Omega037	2026-01-01T04:38:20+00:00	No Flair
2	Google Maps query for whole state	0	3	big_data_mike	2026-01-29T01:38:30+00:00	Projects
3	What is the Best Minecraft Server Hosting?	0	0	GODLIKE-host	None	No Flair
4	Resource: Awesome Marketing Science - A curate...	31	1	Dizzy-Midnight-6929	2026-01-27T21:37:47+00:00	Education
...	...	...	...	...	...	...
1078	Weekly Entering & Transitioning - Thread 03 Fe...	11	39	AutoModerator	2025-02-03T05:01:44+00:00	No Flair
1079	[AI Tools] What AI Tools do you use as a copil...	70	38	limedove	2025-02-02T10:16:24+00:00	Tools
1080	any one here built a recommender system before...	4	11	Emotional-Rhubarb725	2025-02-02T19:59:00+00:00	Projects
1081	Use LLMs like scikit-learn	129	5	No_Information6299	2025-02-01T16:46:39+00:00	Projects
1082	deepseek.com is down constantly. Alternatives ...	0	14	mehul_gupta1997	2025-02-02T14:55:58+00:00	AI

After scrapping data collection , It has 1083 row and 6 columns.

4.2 Description of Attributes

The dataset includes the following attributes:

- **Title**: The textual headline of the Reddit post, representing the main topic of discussion.
- **Score**: The net score of the post, calculated as upvotes minus downvotes, indicating community approval.
- **Comments**: The total number of comments received by the post, reflecting user engagement.
- **Author**: The username of the individual who submitted the post.
- **Post_Time**: The date and time when the post was published, used for time-based analysis.

- **Flair**: The category or tag assigned to the post, such as *Discussion*, *Question*, or *Project*.
- **Engagement** (engineered feature): A derived metric computed as the sum of score and comment count to measure overall interaction.

5.Data Preprocessing

```
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Title     1083 non-null   object
 1   Score     1083 non-null   int64
 2   Comments  1083 non-null   int64
 3   Author    1083 non-null   object
 4   Time      984 non-null    object
 5   Flair     1083 non-null   object
dtypes: int64(2), object(4)
memory usage: 50.9+ KB
```

```
df.isnull().sum()

Title          0
Score          0
Comments       0
Author         0
Time          99
Flair          0
dtype: int64
```

We found that 99 missing values of Time.

5.1 Handling Missing Values

> Drop missing values in critical columns.

```python
initial_rows = len(df)
df.dropna(subset=["Title", "Score", "Comments", "Time"], inplace=True)
cleaned_rows = len(df)
print(f"Rows removed due to missing values: {initial_rows - cleaned_rows}")
```

Out put: Rows removed due to missing values: 99

- Find Duplicated and Remove

```python
duplicates = df.duplicated(subset=["Title", "Author", "Time"]).sum()
df.drop_duplicates(subset=["Title", "Author", "Time"], inplace=True)
print(f"Duplicate rows removed: {duplicates}")
```

Out put: Duplicate rows removed: 0

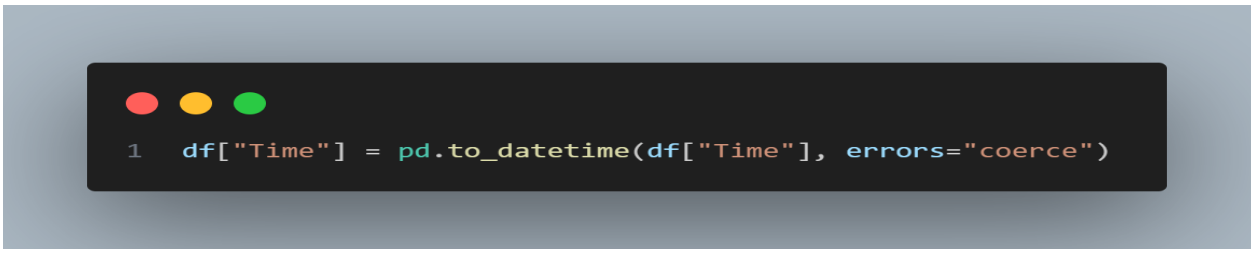
Final Dataset Size: 984 rows

5.2 Data Type Conversion

The Time column is converted from string format into datetime format to allow time-based feature extraction.

Invalid formats are safely handled.

Convert post time to datetime format



```
df["Time"] = pd.to_datetime(df["Time"], errors="coerce")
```

Convert numerical columns to integer



```
df["Score"] = pd.to_numeric(df["Score"], errors="coerce").astype(int)
df["Comments"] = pd.to_numeric(df["Comments"], errors="coerce").astype(int)
```

5.3 Feature Engineering

A new feature called **Engagement** was created by combining the post score and the number of comments. This engineered feature provides a more comprehensive measure of user interaction than individual metrics and supports deeper analysis of post popularity across different flairs and time periods.



```
1  df['engagement'] = df['Score'] + df['Comments']
```

6.Descriptive Statistics

6.1 Summary Statistics (Central Tendency & Dispersion)

	Score	Comments	Title_Length	Comments_per_Score
count	984.000000	984.000000	984.000000	984.000000
mean	103.317073	34.264228	60.077236	2.235517
std	256.030559	45.078560	30.687811	5.667300
min	0.000000	0.000000	10.000000	0.000000
25%	7.000000	9.000000	39.000000	0.302013
50%	26.000000	19.000000	56.000000	0.616922
75%	86.000000	41.000000	72.000000	1.314103
max	2760.000000	433.000000	245.000000	74.000000

The descriptive statistics show that most posts receive low to moderate engagement, while a small number of posts achieve very high scores and comments, indicating a right-skewed distribution.

For Engagement

To gain an initial understanding of the dataset, basic descriptive statistics were computed for key engagement-related variables, including post score, comment count, and the engineered engagement metric. These statistics provide a summary of central tendency and variability and support the visual exploratory analysis.

	Score	Comments	Engagement
count	984.000000	984.000000	984.000000
mean	103.349593	34.265244	137.614837
std	256.130351	45.077844	286.448813
min	0.000000	0.000000	0.000000
25%	7.000000	9.000000	19.000000
50%	26.000000	19.000000	48.000000
75%	85.000000	41.000000	127.000000
max	2761.000000	433.000000	3114.000000

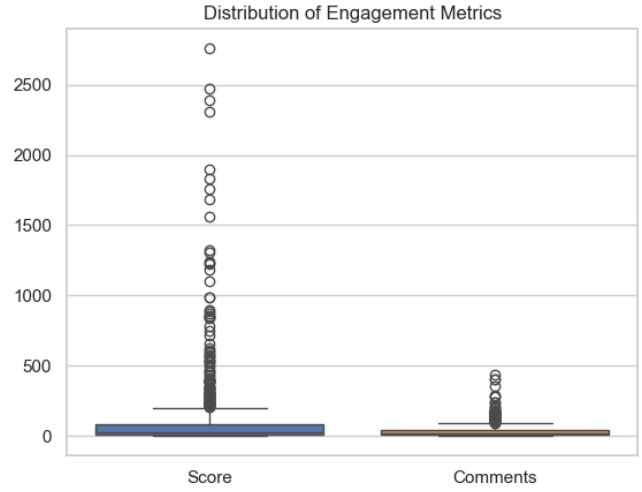
	Q1 (25%)	Median (50%)	Q3 (75%)	IQR
Score	7.0	26.0	85.0	78.0
Comments	9.0	19.0	41.0	32.0
Engagement	19.0	48.0	127.0	108.0

The descriptive statistics indicate substantial variability in engagement-related metrics. In particular, the difference between the mean and median values suggests that the distributions of score, comment count, and engagement are highly skewed. This observation motivates the use of robust descriptive measures, such as the median and interquartile range, in the subsequent analysis.

6.2 Robust Descriptive Statistics

	Mean	Median	Variance	Std Dev	Skewness	Kurtosis
Score	103.317073	26.0	65551.647073	256.030559	5.756157	41.282392
Comments	34.264228	19.0	2032.076603	45.078560	3.741323	21.174728

	Q1	Median	Q3	IQR
Score	7.0	26.0	86.0	79.0
Comments	9.0	19.0	41.0	32.0



The median score is significantly lower than the mean, indicating aright-skewed distribution. This suggests that while most posts receivemoderate engagement, a small number of posts achieve exceptionally highscores. The IQR confirms substantial variability in user engagement.

Outliers were identified in engagement metrics such as Score and Comments. However, these values represent genuine high-engagement posts rather than measurement errors. Since Reddit engagement naturally follows a skewed distribution, outliers were retained. To address skewness, robust statistics and non-parametric tests were used in subsequent analysis.

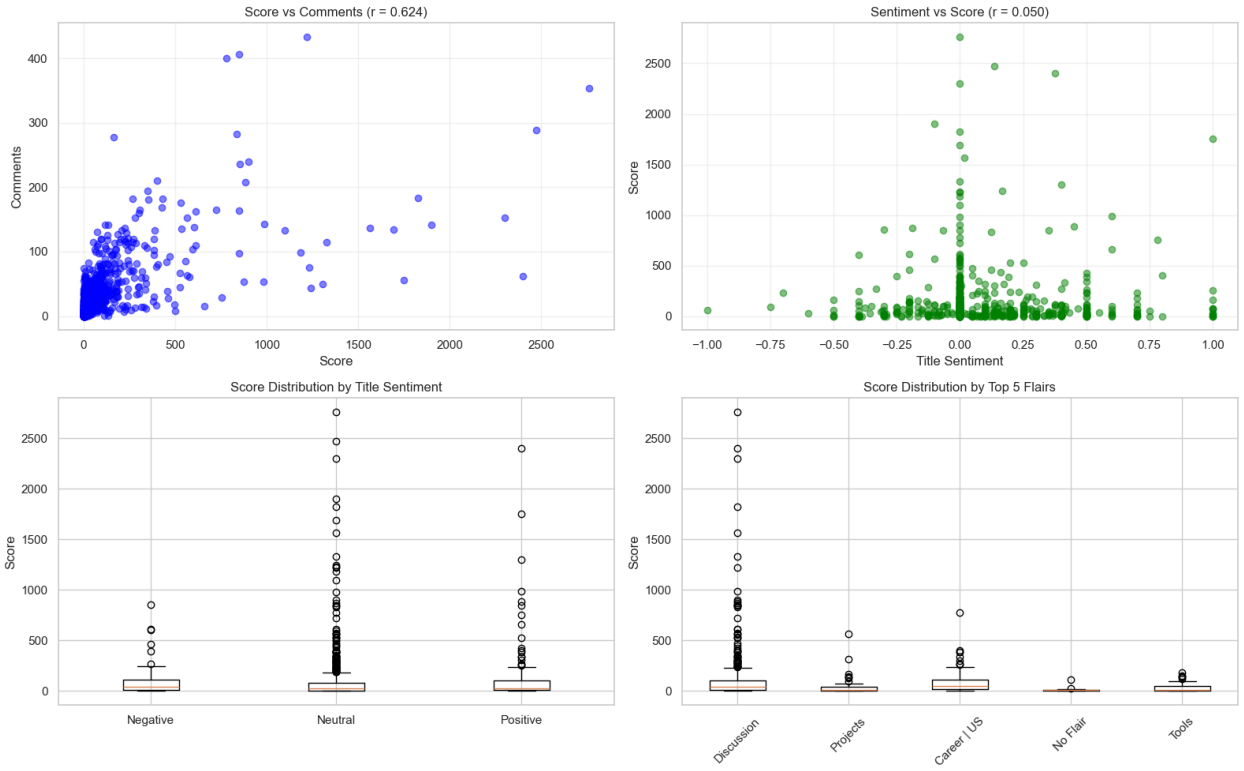
7. Exploratory Data Analysis (EDA)

Time-Series Analysis

Time-based analysis in this project relies on approximate datetime values derived from relative timestamps. While this allows trend analysis, exact posting times may differ slightly from real values.

Engagement relationships

Analyze how different metrics and factors interact to drive user engagement on social media platforms. Our analysis examines four key dimensions:



- **Score vs Comments**: Measures the correlation between post popularity (upvotes/downvotes) and discussion volume. Indicates whether popular posts generate more community interaction.

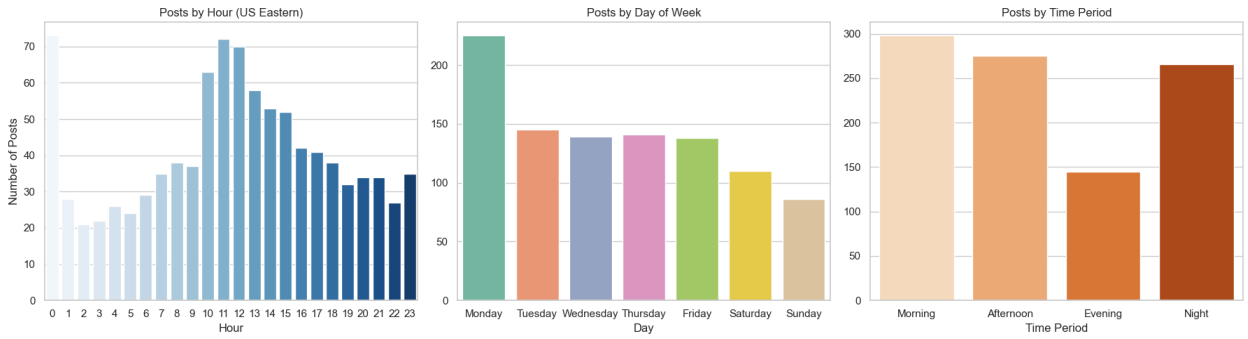
- **Sentiment vs Score**: Shows how the emotional tone of post titles affects engagement. Helps identify whether positive, neutral, or negative titles resonate better with audiences.

- **Sentiment Category Analysis**: Compares engagement levels across sentiment types (Negative, Neutral, Positive) to determine which emotional appeals perform better.

- **Content Type (Flair) Analysis**: Evaluates how different content categories drive engagement, revealing which topics attract more audience attention.

Key Finding: These relationships help identify optimal content strategies for maximizing engagement on social media platforms, particularly relevant for understanding student social media addiction patterns and behavior.

Posting Activity Over Time

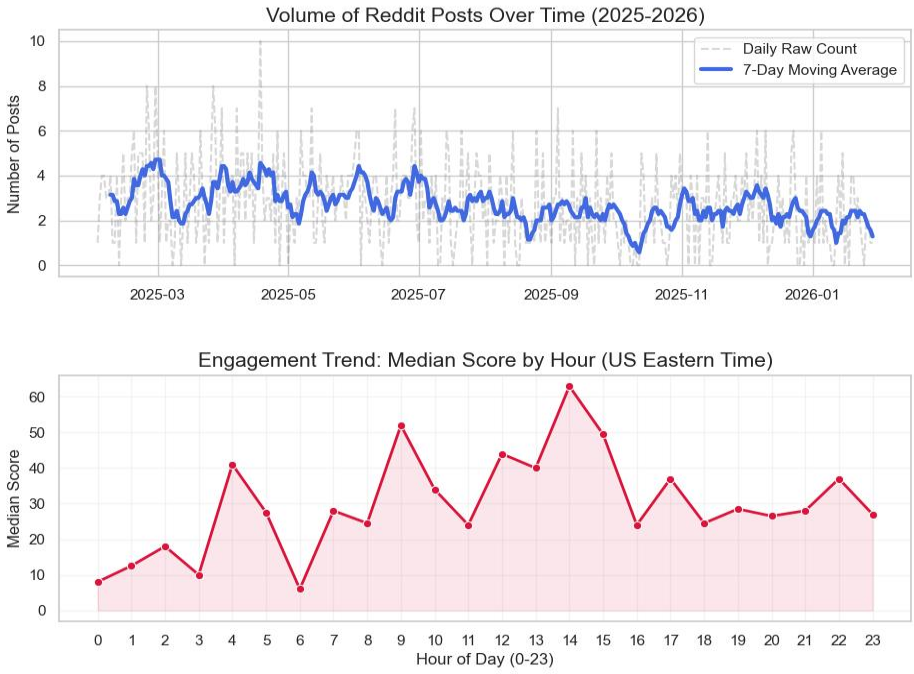


Posting activity is higher on weekdays than weekend, indicating that most Reddit discussions occur during regular work or study days. This pattern suggests that users are more active in engaging with the community while working or studying.

However, **weekend activity remains significant** showing continued user engagement outside weekdays. This demonstrates that the community maintains consistent participation throughout the week, with users also dedicating time to discussions during their leisure time on weekends.

Key Insight: The data reveals a balanced engagement pattern where weekday activity dominates but doesn't eliminate weekend participation, indicating a healthy and consistent user base that maintains regular interaction regardless of the day of the week.

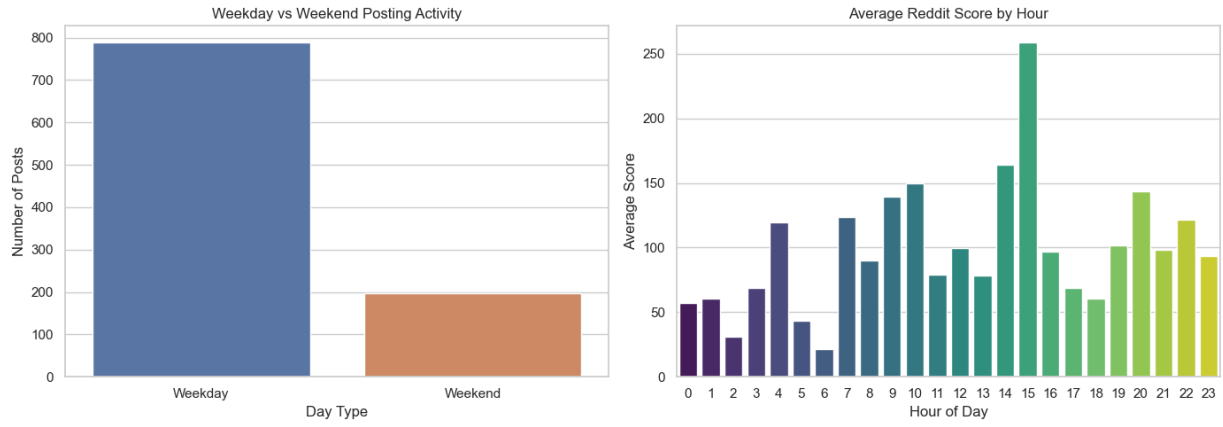
Posts Per Day with 7-Day Rolling Average & Engagement by Hour



Posts Per Day with 7-Day Rolling Average: The 7-day rolling average smooths daily fluctuations to reveal underlying trends in posting volume. Raw daily counts show volatility due to normal variations, but the moving average illustrates the overall trajectory of community activity. Identifying growth, decline, or stability patterns helps understand long-term engagement trends.

Engagement by Hour – Report: Median score analysis by hour reveals when posts achieve peak visibility. The hourly trend identifies optimal posting times when the audience is most active and engaged. Posts published during high-engagement hours tend to accumulate more upvotes and interaction, making timing a critical factor in content success. The fill shading between the trend line and axis emphasizes the engagement intensity throughout the 24-hour cycle.

Weekday vs Weekend + Engagement



Posting Activity Comparison: The left chart reveals the distribution of posts between weekdays and weekends. Weekdays consistently generate more posts, reflecting user behavior during work and study hours when engagement with data science discussions is highest. Weekend posting activity, while lower in volume, remains substantial, indicating sustained community interest outside traditional work hours.

Engagement by Hour: The right chart displays average Reddit scores across all 24 hours of the day. Clear patterns emerge showing:

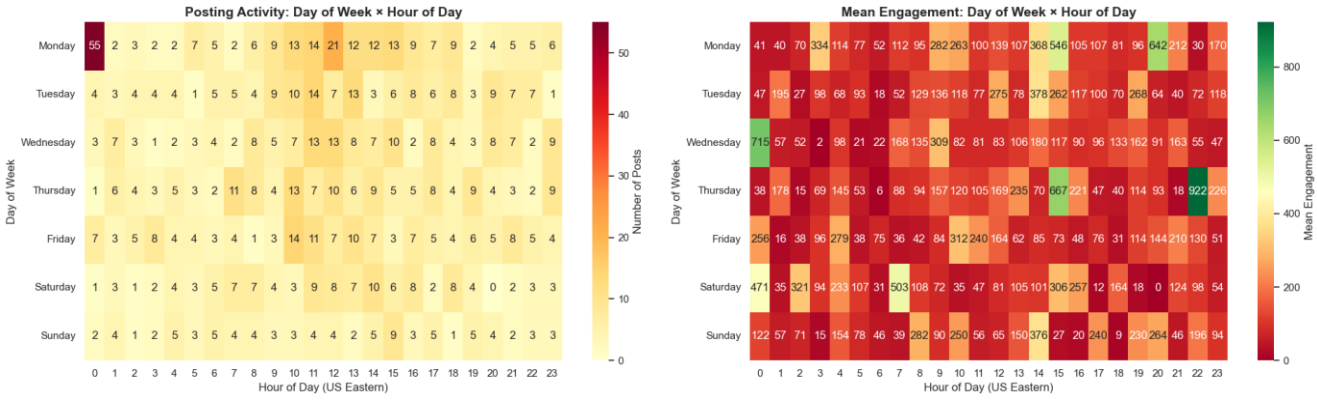
- **Peak Hours**: Afternoon and evening hours (12:00-21:00 ET) show elevated engagement levels

- **Low Hours**: Late night and early morning (0:00-5:00 ET) show minimal engagement

- **Consistency**: Post timing significantly influences visibility and upvotes

Key Insights: Users should post during afternoon/evening hours for maximum engagement. The disparity between weekday and weekend posting reflects the platform's demographics—primarily professionals and students who engage during structured time periods. However, quality content maintains engagement even during off-peak hours, suggesting that content relevance matters alongside timing strategy.

Day of Week × Hour of Day Heatmaps

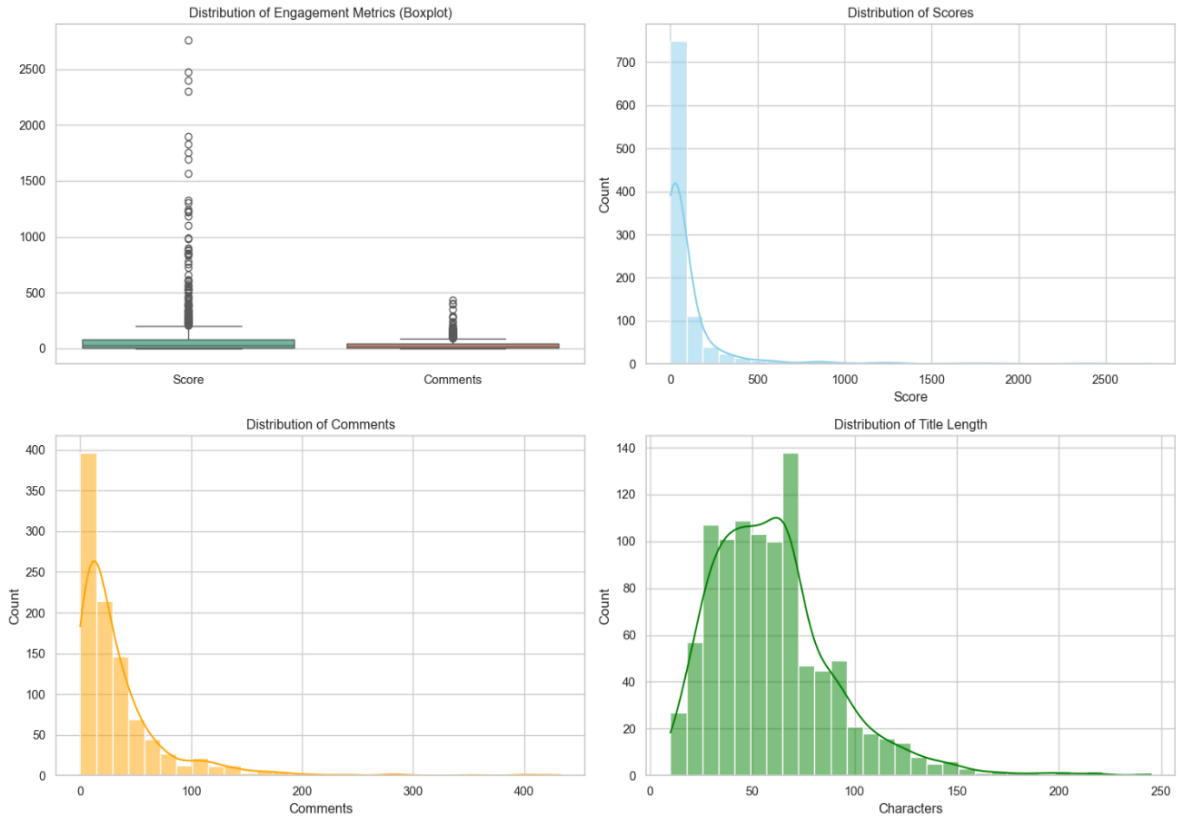


Heatmap of Posting Activity by Day and Hour: The heatmap of posting activity shows that Reddit posts in the r/datascience subreddit are not evenly distributed across time. Posting frequency tends to be higher during weekdays compared to weekends, with noticeable peaks around late morning to early afternoon hours (approximately 9:00–14:00 US Eastern). In contrast, weekend activity appears lower and more evenly spread throughout the day. This pattern suggests that users are more active during typical working or study hours.

Heatmap of Mean Engagement by Day and Hour: The engagement heatmap reveals that high engagement does not always coincide with periods of high posting activity. Certain time slots exhibit relatively low posting frequency but high average engagement, indicating that posting

volume alone does not determine interaction levels. Engagement varies substantially across both day of week and hour of day, highlighting the influence of temporal factors such as visibility and audience availability.

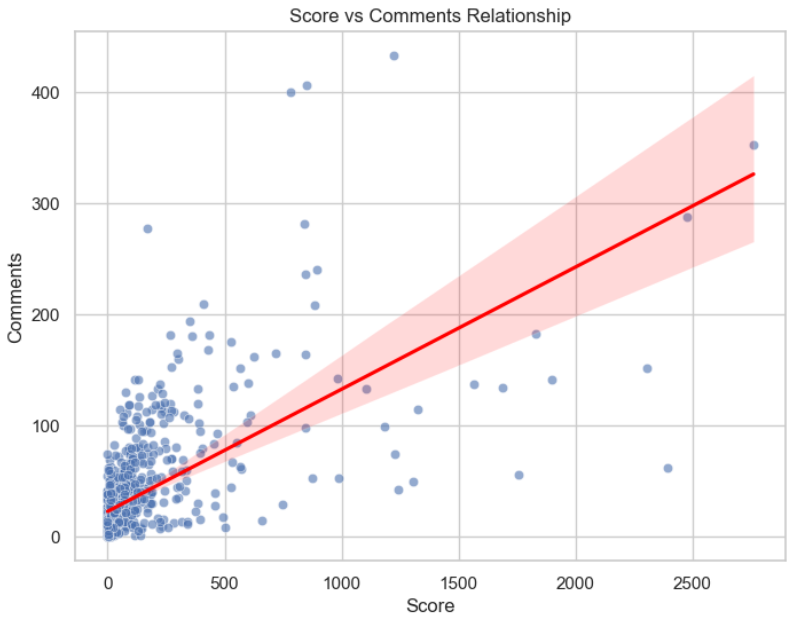
Engagement Distributions



Distribution Patterns: Score and Comments both exhibit strong right-skew — most posts cluster at low values while a small number achieve extremely high engagement, confirming the "viral outlier" phenomenon. Title length shows a more normal distribution, suggesting posting length doesn't strongly constrain visibility.

Key Insight: The extreme right skew means median engagement is far below the mean, indicating that consistent, reliable engagement is modest. However, those who optimize timing and content category can break into the viral tail. Length moderation appears acceptable—extremes in either direction don't harm engagement.

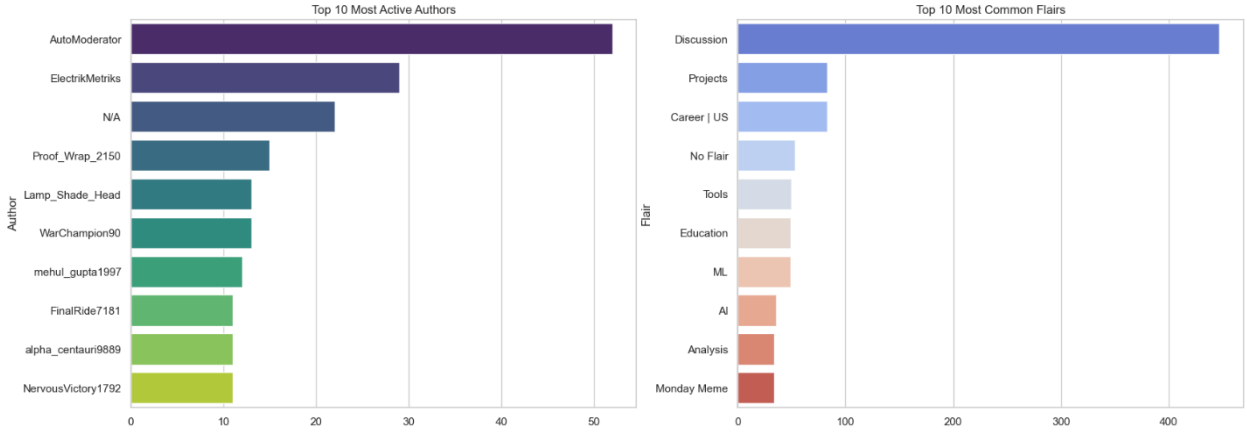
Relationship Between Score and Comments



Score–Comment Relationship: The scatter plot reveals a moderate positive correlation ($r \approx 0.6$–0.7), indicating that high-scoring posts tend to attract more discussion. However, substantial scatter around the trend line shows the relationship is imperfect—some posts gain upvotes without comments (passive endorsement) while others spark discussion despite modest scores (niche appeal).

Key Insight: Visibility and discussion are linked but independent. Maximize both by targeting both upvote-friendly (timely, clear) and discussion-provoking (novel, controversial) content. The loose coupling suggests quality arguments can engage even without mass appeal.

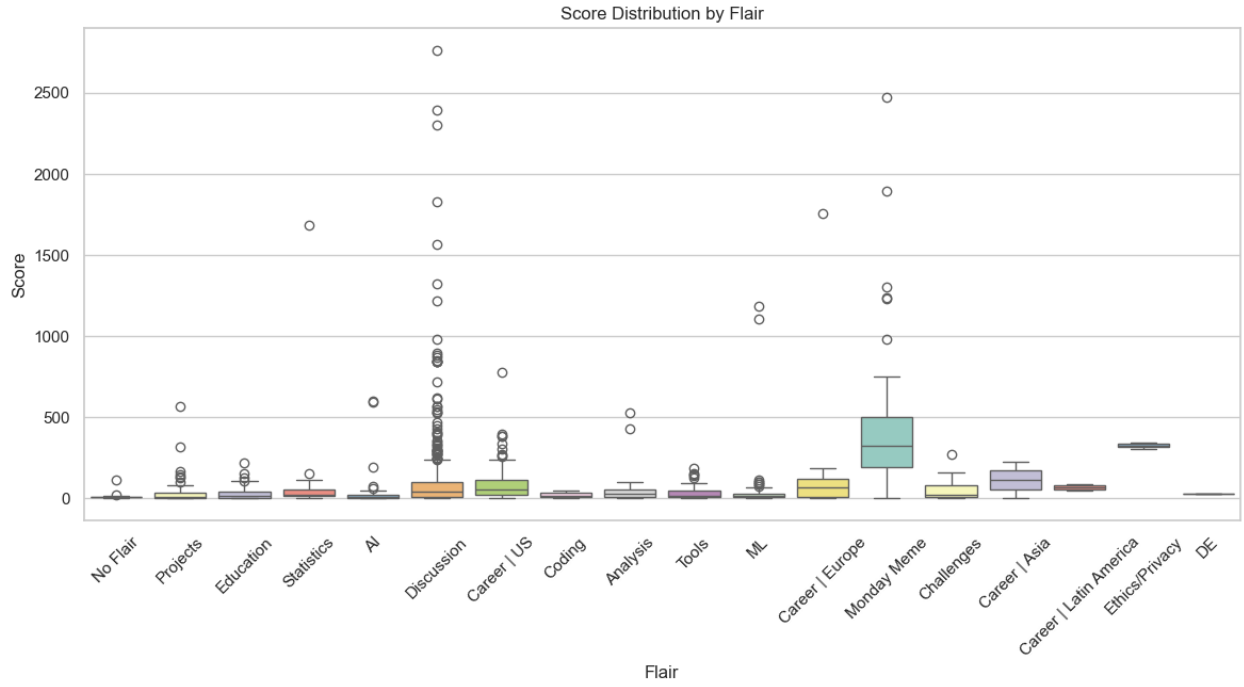
Top 10 Authors & Flairs



Community Structure: The left chart shows a small cohort of prolific authors dominate r/datascience, with the top contributor posting significantly more than others. The right chart reveals flair distribution is also concentrated—a few categories (e.g., "Discussion," "News") generate most posts, while many niche flairs appear infrequently.

Key Insight: Content is authored and categorized by a power-law distribution. Repeat contributors build authority; strategic use of popular flairs increases discoverability. However, niche flairs serve engaged micro-communities, so specialized content finds its audience despite lower volume.

Score by Flair (Boxplot)

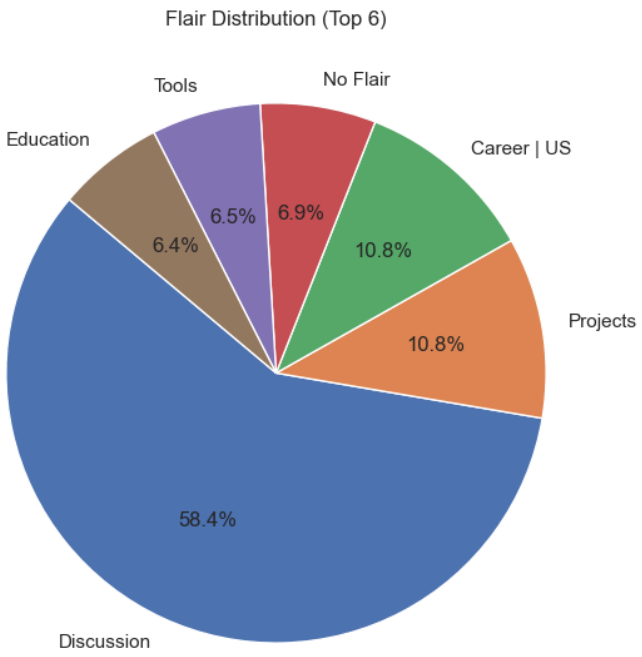


Flair–Engagement Variance: The boxplot reveals that certain flairs (e.g., "Discussion," "Research") consistently achieve higher median and maximum scores, while others remain confined to lower ranges. Flairs with tighter boxes show consistent engagement; those with long whiskers indicate occasional viral spikes within that category.

Key Insight: Content category is a strong engagement driver—choose flairs associated with higher median scores. However, variability within each flair suggests that execution (timing, title clarity, relevance) still matters. Even low-median flairs can produce successful posts with strong fundamentals.

pie chart: Top 5 flairs + Others grouped

The pie chart shows the relative share of posts from the top five flairs, with all remaining flairs grouped into "Other." It highlights which flairs dominate posting volume and how much of the dataset comes from the long tail of less-common flairs. Use this to focus analysis or recommendations on the most frequent flairs while acknowledging the combined impact of many smaller categories.



8. Key Findings

Based on the exploratory data analysis of posts from the r/datascience subreddit, several key findings were identified. Posting activity varies over time, indicating fluctuating levels of community participation. Engagement metrics such as post scores and comment counts are highly right-skewed, with most posts receiving low interaction and a small number of posts attracting very high engagement. Analysis of the relationship between scores and comments shows a positive association, suggesting that posts receiving more upvotes tend to generate more discussion. Additionally, flair-based analysis reveals that discussion- and career-related posts generally achieve higher engagement compared to other categories. Overall, the results highlight uneven attention distribution and distinct engagement patterns within the community.

Additionally, temporal analysis using hourly and weekly heatmaps shows that both posting activity and engagement vary significantly by time of day and day of week, with engagement peaks not always aligning with periods of high posting activity.

9. Limitations of the Study

This study has several limitations. The dataset represents a limited time period and may not reflect long-term trends in subreddit activity. The analysis focuses only on post-level metadata and does not include the textual content of posts or comments. Data collection was performed through web scraping rather than the official Reddit API, which may limit data completeness. Additionally, engagement metrics such as scores and comments can be influenced by external factors such as post timing and visibility. Therefore, the findings are specific to the r/datascience subreddit and may not be generalizable to other communities.

Time-based patterns were analyzed descriptively and do not account for external factors such as post visibility algorithms or user time zones.

10. Conclusion

This project demonstrated the use of web scraping and exploratory data analysis to study posting behavior and engagement patterns in the r/datascience subreddit. Using Python, a dataset of approximately 1,100 Reddit posts was collected, cleaned, and analyzed to identify trends and relationships within the data. The analysis showed that user engagement is highly uneven, with a small number of posts accounting for a large share of interactions, and that certain types of content attract higher engagement. By focusing on exploratory analysis rather than inferential or predictive modeling, the study provides a clear overview of community dynamics while remaining aligned with the project objectives. The findings illustrate the value of social media data for understanding online discussion behavior.

Temporal exploratory analysis further demonstrated that posting behavior and engagement in the r/datascience subreddit are influenced by time-related factors, highlighting the importance of when posts are published.

11.Future Work

This project can be extended in several ways to provide deeper insights into the r/datascience community. Future work could include analyzing the textual content of posts and comments using natural language processing techniques to identify common topics and sentiment patterns. Additional engagement metrics, such as comment depth or post visibility over time, could also be explored. Collecting data over a longer time period would allow for more robust time-series analysis and trend detection. Finally, using Reddit's official API could improve data completeness and reliability compared to web scraping.

12.References

Website: https://old.reddit.com/r/datascience